

Yixing Wu

Los Angeles, CA | yixingwu98@gmail.com | +1 (323) 818-9613 | star-wyx.github.io/
linkedin.com/in/yixing-wu-52811a169 | github.com/star-wyx

Summary

Ph.D. student in Electrical Engineering at USC, focusing on **Generative AI**, **Computer Vision**, and **efficient machine learning**. Experience with **Transformer-based models** and applied ML systems in research settings. Seeking **Machine Learning Engineer / Applied Scientist** internships.

Research Interests

Generative AI & Computer Vision: Generative modeling for image restoration, super-resolution, and image understanding.

Efficient Machine Learning: Designing compact and scalable models for high-dimensional visual and time-series data, with an emphasis on robustness and practical deployment.

Education

University of Southern California Ph.D. Student in Electrical Engineering, Minor area in Computer Science Advisor: Professor C.-C. Jay Kuo	Los Angeles, United States 08/2024 - Present
University of Southern California M.S. in Electrical Engineering (Honors)	Los Angeles, United States 01/2022 - 12/2023
Xidian University B.S. in Information Engineering	Xi'an, China 09/2016 - 07/2020

Selected Publications (Full list at: [\[Google Scholar\]](#))

- [1] **Yixing Wu**, Vinay Duddalwar, Manju Aron, C.-C. Jay Kuo, "IHC Image Quantification with Green U-Shaped Learning," *Submitted to IEEE International Conference on Image Processing (ICIP)*, 2026.
- [2] Wei Wang, **Yixing Wu**, C.-C. Jay Kuo, "A Green Learning Approach to LDCT Image Restoration," *2025 IEEE International Conference on Image Processing (ICIP)*.
- [3] **Yixing Wu**, A. Shieh, S. Cen, D. Hwang, X. Lei, S. J. Pawan, M. Aron, I. Gill, C.-C. Jay Kuo, "Prediction of PD-L1 and CD68 in Clear Cell Renal Cell Carcinoma with Green Learning," *Journal of Imaging*, 2025.
- [4] Dingyi Nie, **Yixing Wu**, C.-C. Jay Kuo, "Modeling Irregular Multivariate Time Series with Missing Observations," *APSIPA Transactions on Signal and Information Processing*, *Accepted*.
- [5] Xinyu Wang, **Yixing Wu**, Haiyi Li, Vinod K. Mishra, C.-C. Jay Kuo, "A Statistics-based Feature Generation (SFG) Method: Theory and Applications," *2024 IEEE International Conference on Big Data (BigData)*.

Awards

USC Viterbi Research Assistantship	08/2024 - Present
USC EE MS honors program	09/2023 - 12/2023

Selected Research Projects

Diffusion-based Generative Modeling for Image Restoration • Developing DDPM/DDIM-based diffusion frameworks for image denoising and inverse problems. • Exploring conditioning mechanisms and noise scheduling strategies for stable training under limited data. • Analyzing computational efficiency and robustness compared to standard feedforward generative models.	09/2025 – Present
Transformer-based Modeling and Efficient Inference • Implemented and experimented with decoder-only Transformer architectures in research settings. • Explored multi-GPU training with PyTorch Distributed Data Parallel and studied memory-efficient inference	10/2025 – Present

strategies.

- Investigated scaling behavior and latency–memory trade-offs in long-context modeling.

Efficient Multi-Stage Vision Modeling for Segmentation and Classification

07/2025 – 01/2026

- Proposed a multi-stage, coarse-to-fine framework for segmentation and instance-level classification under multimodal supervision.
- Designed a feedforward, multi-resolution architecture achieving competitive accuracy with significantly reduced FLOPs and model complexity.

Generative Modeling for Image Restoration and Inverse Problems

09/2024 – 01/2025

- Developed parameter-efficient generative models for image denoising under severe noise and limited data.
- Designed multi-stage architectures to enhance robustness and stability under distribution shift.
- Achieved competitive performance with reduced computational and memory cost.

Hierarchical Modeling for Long Time-Series Data

06/2025 – 09/2025

- Developed hierarchical and multi-resolution models for long and noisy time-series analysis.
- Designed coarse-to-fine modeling strategies to improve stability and computational efficiency in long-sequence learning.

Technologies

Programming Languages: Python, C/C++, Java, MATLAB

ML Frameworks: PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost, OpenCV

Systems: Linux, Git, Docker, AWS (EC2, Lambda)